

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Science of Living Systems (LS1001)

Time: 1.5 Hours

Semester / Programme: 1st Semester B.Tech

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) What structural feature of membrane phospholipids prevents close packing of fatty acid tails, thereby maintaining plasma membrane fluidity at lower environmental temperatures? [1]

1(b) Distinguish between totipotent and pluripotent stem cells in terms of their developmental differentiation potency. [1]

1(c) Name the specific catalytic ribosomal component that functions as the peptidyl transferase enzyme during polypeptide chain elongation in prokaryotic protein synthesis. [1]

1(d) State how the kinetic parameters V and K_m are affected in the presence of a competitive enzyme inhibitor. [1]
 $K_m^{\text{app}} = K_m \left(1 + \frac{[I]}{K_i} \right), \quad V_{\text{max}}^{\text{app}} = V_{\text{max}}$

1(e) Identify the primary electron donor and the ultimate electron acceptor of non-cyclic electron transport (Z-scheme) in oxygenic photosynthesis. [1]

Formula: $2\text{H}_2\text{O} \xrightarrow{h\nu} \text{O}_2 + 4\text{H}^+ + 4e^-$, $\text{NADP}^+ + 2e^- + \text{H}^+ \xrightarrow{\text{FNR}} \text{NADPH}$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Tabulate three fundamental structural differences between prokaryotic and eukaryotic ribosomes. Explain how these distinctions form the molecular basis for the selective toxicity of broad-spectrum aminoglycoside or macrolide antibiotics. [2.5]

(b) Define cellular homeostasis. Illustrate and describe the essential components of a negative feedback control loop (sensor, control center, effector) taking blood glucose regulation in humans as an engineering control systems analogue. [2.5]

Q.3. (a) Explain the four structural levels of proteins (primary, secondary, tertiary, and quaternary). Detail the specific chemical bonds and interactions that drive and stabilize the tertiary structure of a globular protein. [3]

(b) What are Induced Pluripotent Stem Cells (iPSCs)? State any two transcription factors discovered by Shinya Yamanaka used for cellular reprogramming, and discuss one biomedical application of iPSC technology. [2]

- Q.4. (a)** Given the Michaelis-Menten kinetic expression $v_0 = (V[S] / K_m + [S])$, mathematically derive the Lineweaver-Burk double-reciprocal equation. Explain how the kinetic constants V and K_m are experimentally extracted from the slope, y-intercept, and x-intercept of the resulting plot. [3]

$$\frac{1}{v_0} = \left(\frac{K_m}{V_{\max}} \right) \frac{1}{[S]} + \frac{1}{V_{\max}}$$

- (b)** Describe four distinct catalytic strategies used by enzymes to accelerate reaction rates by lowering activation energy (E_a). [2]

- Q.5. (a)** Outline the path of electron transport in the photosynthetic Z-scheme from Photosystem II (PSII) to Photosystem I (PSI). Explain how this electron flow couples with proton translocation to drive ATP synthesis via Mitchell's chemiosmotic hypothesis. [3]

Formula: $\text{ADP} + \text{P}_i + n\text{H}^+_{\text{lumen}} \xrightarrow{\text{ATP Synthase}} \text{ATP} + n\text{H}^+_{\text{stroma}}$

- (b)** Summarize the three sequential stages of the Calvin Cycle (Dark Reaction) in C_3 photosynthesis. Write the net stoichiometric equation showing the input of CO_2 , ATP, and NADPH required to synthesize one molecule of glucose. [2]



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